

Inferring Urban Morphology through Gravitational Work Minimization of Twitter-Derived Trajectories

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Abstract

INTRODUCTION

METHODS

Twitter OD Data Reconstruction

Next, we confronted the challenge of separating genuine human movement from automation, one-off events, and artifacts. We begun by identifying accounts whose activity pace exceeded three posts per hour; such high-frequency behavior almost always reflects bots, live-event streams, or institutional feeds rather than individual movements. Once those accounts were removed, we discarded any remaining users with fewer than three total posts, since a single hop or a pair of points cannot trace a meaningful path. Finally, even plausible streams could hide implausible leaps: we computed, for each consecutive pair of posts, the straight-line distance in meters and the elapsed time in hours, derive an implied velocity, and eliminate any segments exceeding 900 km/h: this approach filtered out multi-account profiles—often corporate or organizational actors—ensuring our analysis focused solely on individual users.

This three-stage cleansing left us with a curated collection of trajectories that credibly represent human mobility.

Having distilled the data to its essence, we superimposed a regular, square grids—of side length 1000 m—over the projected extent of each city. Then we computed the overall bounding box of the reprojected tweets and subdivided it into integer-indexed cells, yielding a uniform lattice that simplifies both analysis and interpretation. An efficient spatial index then assigned each tweet to its corresponding cell with the following logic: precise POINT geometries mapped directly via integer division, while POLYGON footprints were first intersected with the grid and then resolved—by selecting the cell of maximum overlap. In this way, every tweet had been endowed with a unique cell identifier and an associated confidence score, enabling consistent, multiscale trajectory analysis.

The final transformation converted these cell-tagged trajectories into origin-destination flows. We label each date as a working day or holiday—either by consulting a calendar API—and then, for each user, chronologically sequence their visited cells. Every successive pair of cells contributed one unit to the corresponding entry in an OD matrix. Summing over all users and days of the same type yielded two comprehensive flow tables—one for workdays and one for holidays—where each row specified an origin cell, a destination cell, and the total count of observed transitions. These OD matrices form the empirical backbone of our gravitational-work minimization, enabling us to probe how subtle perturbations to the urban fabric—through translations, rotations, or scalings—reshape the uphill effort embedded in real human journeys. In Figure 1 we show an example of the OD matrix flows reconstructed for the city of Barcelona for the working days (left panel a) and holidays (right panel b).

Study Areas and City Boundaries

For this study, we selected a set of cities that represent different urban landscapes and mobility patterns. The cities included in our analysis are shown in Figure 2 and include Rome (Italy), Madrid (Spain), Barcelona (Spain), Boston (USA) and Santiago del Chile (Chile)..

The geographic boundaries of each city were defined using the spatial dataset delineates the boundaries of Functional Urban Areas (FUA) of Urban Centres, which was obtained from the European Commission’s Joint Research Centre (JRC) [1]. This dataset provides a comprehensive and standardized definition of urban areas across Europe and beyond, allowing for consistent comparisons of urban morphology and mobility patterns. In the Supplementary Material (see Appendix ??), we provide a table summarizing the key characteristics of each city, including population size, area, and the number of Twitter-derived trajectories analyzed. Moreover, we show the boundaries of the FUA for each city in Figure ??, which illustrates the spatial extent of the area as defined by the FUA dataset. Each figure represents the functional connections between the urban core and surrounding areas, which are crucial for understanding urban morphology and mobility patterns.

The FUA boundaries were used to extract the relevant urban network data from OpenStreetMap (OSM) and to define the spatial extent for the analysis of Twitter-

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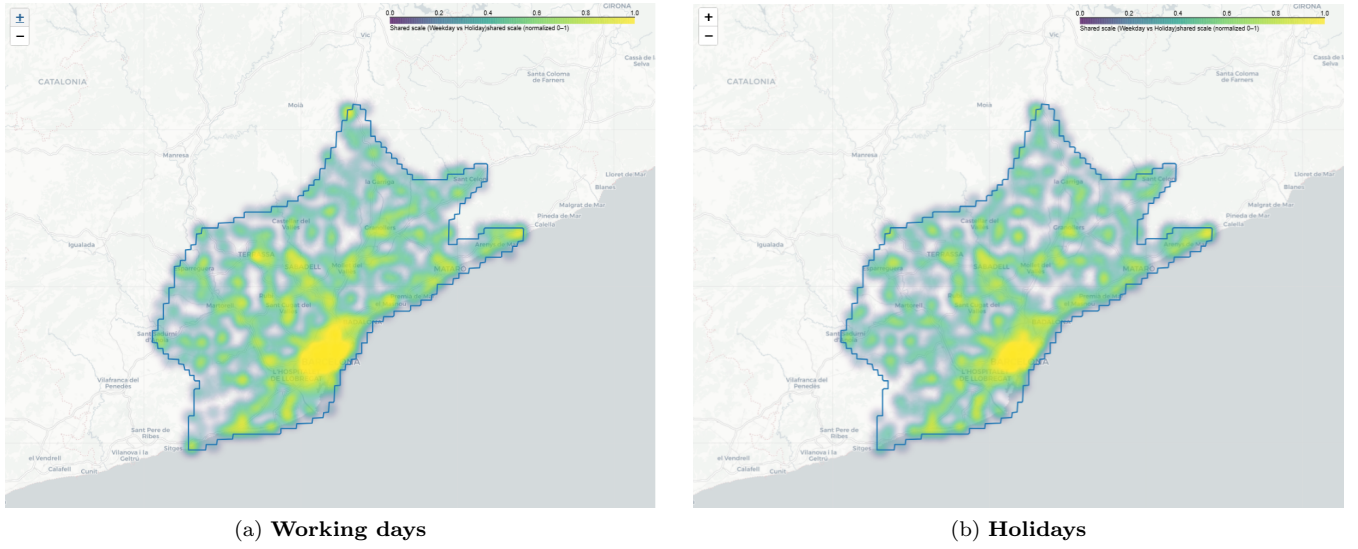


FIG. 1. **Origin-destination flow heatmaps for Barcelona.** a–b, Working days (a) and holidays (b) OD flow heatmaps, illustrating the spatial distribution of user movements across the city. The color intensity of each cell corresponds to the normalized number of transitions between origin-destination pairs, with a color bar indicating the scale. The axes represent the grid cell indices along the x and y directions, respectively.

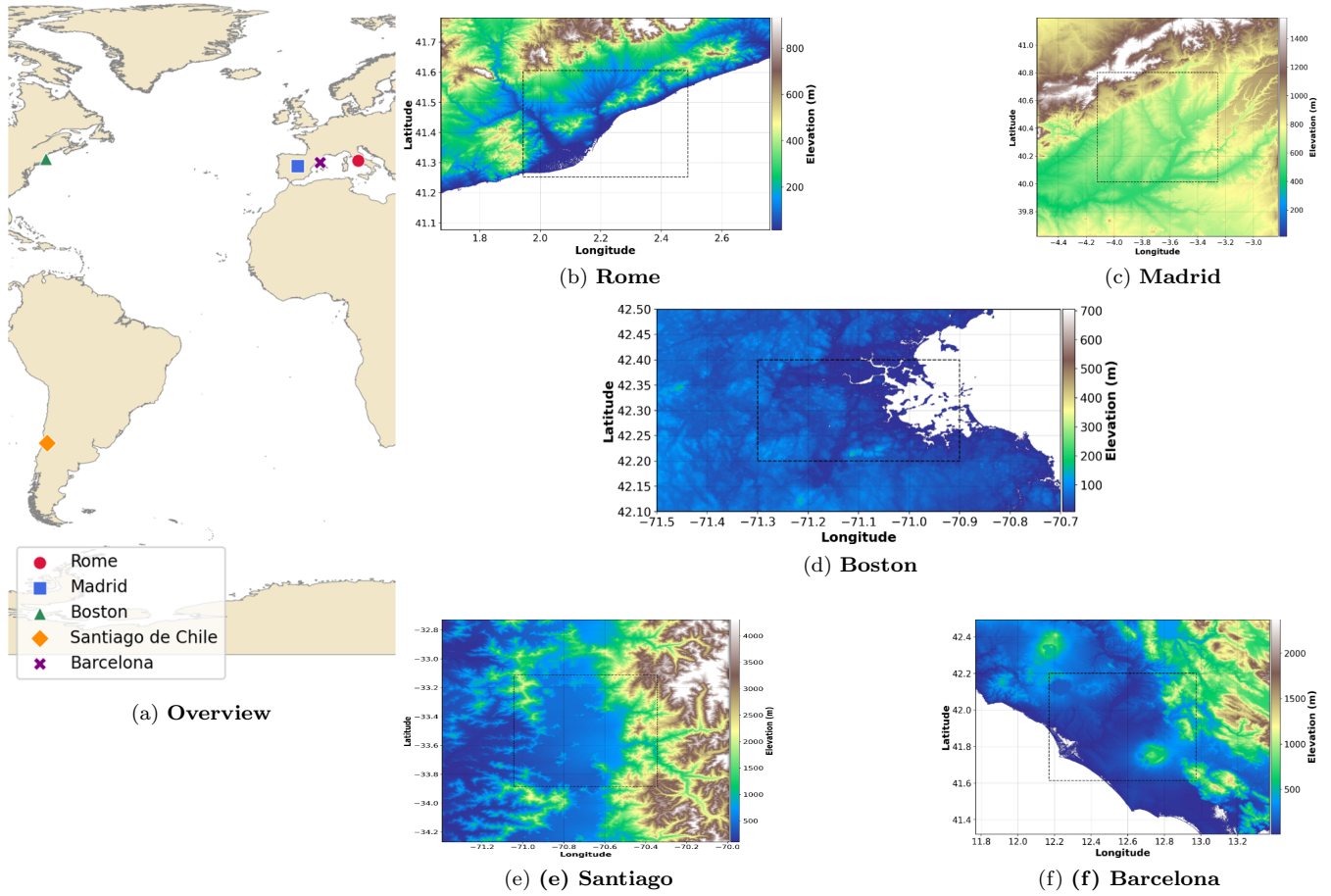


FIG. 2. **Topography of selected cities.** Global overview on the left (a); local maps on the right in a 2+1+2 layout (b–f).

derived trajectories. The FUA boundaries were reprojected to match the coordinate system of the Twitter-

FIG. 3. **London Heathrow Airport aggregated copresence network of one month data.** a Sketch ...

derived trajectories, ensuring that all spatial operations were consistent.

Gravitational Work Computation

To quantify the topographic cost of movement, we first acquired a high-resolution SRTM digital elevation model (DEM) for each city, as described in Sec.

At this point, we discretized each road-segment into evenly spaced sample points at a fixed interval $\Delta s = 10$ m. For a given edge e , of the graph G , of total length L_e , we set the total number of elevation-sampling points placed along edge e as

$$N_e = \left\lceil \frac{L_e}{\Delta s} \right\rceil + 1,$$

and defined the sampling distances

$$s_i = \frac{i}{N_e - 1} L_e, \quad i = 0, 1, \dots, N_e - 1.$$

Each point $\mathbf{x}_i$ was obtained by interpolating the linestring geometry of e at distance s_i , then projected to geographic coordinates via the DEM's affine transform. We queried the DEM at $\mathbf{x}_i$ (using rasterio's sampling routine) to retrieve an elevation value z_i . The resulting elevation profile for edge e is therefore the ordered sequence

$$\{(\mathbf{x}_i, z_i)\}_{i=0}^{N_e-1},$$

which provided the detailed vertical trace needed for accurate uphill-work computations.

Copresence dynamic network

A first question to address is

EBlablaba

The contact network is

Spatial Immunization

The diseases mentioned

Name	Explanation	Reference
SIR	Original SIR	[?]
SEIR	SEIR for H1N1	[?]
EBOLA	SEIR for EBOLA	[?]
SARS-Cov2:	SEIIR for SARS-Cov2	[?]
SIRR3	SIR double infectivity	[?]
SIR30	SIR with 30' time scale	[?]

TABLE I. List of abbreviation for the epidemic models here considered and respective reference to the literature for the choice of the epidemic parameters.

FIG. 4. **Risk of contagion in space.** Cells in the airport area with a heatmap showing the fraction of contagions occurring in them when running the SIR model. In **a**, after one day of simulation and in **b** after two days.

RESULTS

Simplest model: SIR

We start by studying the simplest epidemic model: the SIR, with only susceptible, ...

SEIR family

DISCUSSION AND CONCLUSIONS

In this work, we have shown how using a network approach it is possible to reduce the complexity of the spatial trajectory in Heathrow airport and encode them in a temporal network of copresences among different categories of users. Based on those networks, we run four different compartmental models from the literature to mimic the spreading of four different strains, namely SARS, H1N1, EBOLA and SARS-Cov2, each one with different epidemic parameters. From these simulations we are able to detect workers as the most exposed and infectious category of individuals inside the airport, independently of the virus under study. This is due to the fact that employees are the agents more often infected: they recurrently appear in the airport day after day and they come in contact with many passengers, but also colleagues, whereas passengers generally cross the area of the airport only for few hours before departure or after arrival. To fully describe the movements across the airport, we had to supersample the trajectories creating a standard day which is of course only an approximation of reality, but still these different behaviors hold true in our simulations. Second in the rank by infectiousness we find connecting passengers, since they are the travelers who spend more time inside the airport waiting for connections between a flight and the next. Restaurants, bars and in general relax areas are the places where most

of the copresences (and contagions) happen and this is where indeed connecting travelers use to rest and stop for a long time waiting and interacting with employees.

By applying a policy of spatial immunization based on non-harmful UV lights or similar spatial immunization methods such as frequent cleaning and disinfection of surfaces, air filters, ozone, etc, to hygienize the area of terminals, we are able to decrease the effect of infectious droplets, aerosols and contaminated air or surfaces. By selecting the places to immunize and their number using infection hotspots as predicted by the models, it is possible to sensibly decrease the contagion probability and the intensity of outbreaks for every disease studied. In all of them there is a common attribute and it is that the characteristic times of the disease evolution, e.g., latency period, are longer than the period spent by most of the users in the airport. This makes possible that the simplest SIR model, in which only contagion events are relevant, is enough to produce an accurate estimation of the contagion hotspots valid for most diseases. Note that these hotspots are not only characterized by the concentration of people but by a balance between time spent and individual concentration: places as cafeterias, access points, etc, are the most critical hotspots to immunize. Still, the method is general and in the eventuality of considering a quick evolving disease the hotspots should be recalculated with an *ad hoc* model.

We include an analysis further disaggregating the average probability of traveling from each terminal of Heathrow to UK, European and Intercontinental destinations around the world. Thanks to this detailed information, we are able to estimate the average threat represented by the development of an outbreak in the airport for these three classes of destinations for any of the four types of virus. Our spatial immunization policy helps containing the menace of passengers getting infected in the terminals while traveling to other destinations, potentially delaying the massive seeding of new infected individuals from London to the rest of the world. Unfortunately, this policy seems to be less effective on workers,

who exhibit a longer exposure to the virus with respect to passengers. This suggests that targeted vaccinations on the airport employees may further reduce the probability and even the severity of an outbreak within every strain scenario. However, when vaccination is not available, in all the cases we register an important decrease on the outbreak probability and on the total amount of infected among all the destinations and all the categories of individuals. This same approach finds immediate direct applications of a similar spatial immunization technique to many other buildings like hospitals, schools, universities, train stations and museums, potentially reducing the appearance of new and uncontrolled outbreaks even in the absence of vaccines. All these places share the common feature of having a large in and outflow of visitors and by a set of workers that appear in some areas of the infrastructure recurrently. This leads to strong heterogeneities in the location and duration of the contacts among visitors and between visitors and workers, which allows us to define a successful spatial immunization strategy based on a hotspot analysis.

DATA AVAILABILITY STATEMENT

Mobility data is available from Cuebiq through their Data for Good program (<https://www.cuebiq.com/about/data-for-good/>). Restrictions apply to the availability of this data, which was used under license for the current study.

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AUTHOR CONTRIBUTION

To decide

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